

VERSAILLES 3 NNW, KY, USA

WMO: 724300

Lat: 38.0945N

Lon: 84.7465W

Elev: 891

StdP: 14.23

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 63838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.0	(c) 13.3	(d) -1.3	(e) 5.3	(f) 10.5	(g) 3.4	(h) 6.7	(i) 13.4	(j) 21.3	(k) 48.8	(l) 19.8	(m) 46.9	(n) 5.1	(o) N/A	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.2	(c) 90.5	(d) 74.0	(e) 88.0	(f) 73.7	(g) 86.0	(h) 73.0	(i) 77.3	(j) 86.1	(k) 76.0	(l) 84.2	(m) 74.8	(n) 82.6	(o) 6.7	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	134.3	82.8	73.4	128.2	81.1	72.3	123.6	79.5	41.3	86.1	40.0	84.3	38.9	82.6	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.0	(b) 15.5	(c) 13.7	(e) 1.1	(f) 93.3	(g) 6.4	(h) 4.2	(i) -3.5	(j) 96.3	(k) -7.2	(l) 98.7	(m) -10.8	(n) 101.1	(o) -15.4	(p) 104.1		
(a) 18.0	(b) 15.5	(c) 13.7	(e) 0.2	(f) 79.4	(g) 7.3	(h) 1.6	(i) -5.1	(j) 80.6	(k) -9.3	(l) 81.5	(m) -13.4	(n) 82.3	(o) -18.7	(p) 83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	33.0	36.1	45.4	55.9	65.2	72.5	75.0	74.3	68.6	57.1	45.7	37.7
	DBStd	17.40	12.27	11.70	11.27	9.07	7.94	5.11	4.50	4.86	6.82	9.04	9.66	10.11
	HDD50	1852	541	409	227	47	3	0	0	0	0	33	193	399
	HDD65	4557	991	810	610	291	100	7	1	3	41	277	581	845
	CDD50	3913	15	19	85	223	476	676	774	753	557	252	64	19
	CDD65	1144	0	0	4	17	108	233	310	291	148	31	2	0
	CDH74	9065	0	1	18	112	746	1844	2512	2355	1231	240	6	0
	CDH80	2789	0	0	0	4	137	553	843	801	401	50	0	0
	WSAvg	5.9	7.8	7.6	7.7	7.0	5.0	4.9	4.4	4.1	4.4	5.4	6.1	6.8
	PrecAvg	48.5	3.5	3.2	4.5	4.5	5.2	4.5	5.0	3.7	3.5	3.3	3.3	4.3
(15) Precipitation	PrecMax	65.5	5.3	5.6	14.0	13.5	10.6	8.6	9.4	8.5	9.2	6.5	7.8	10.0
	PrecMin	32.6	1.2	1.3	1.6	1.6	1.0	1.1	1.8	1.3	0.7	0.6	1.0	1.7
	PrecStd	7.4	1.1	1.3	2.4	2.5	2.4	1.8	2.2	1.7	2.1	1.9	1.6	1.8
	PrecStd	7.4	1.1	1.3	2.4	2.5	2.4	1.8	2.2	1.7	2.1	1.9	1.6	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.9	68.9	76.5	80.5	86.5	91.1	93.9	94.1	92.6	84.8	74.0	66.2
		MCWB	58.3	59.8	64.1	64.0	72.3	73.5	73.8	76.4	71.2	68.0	62.1	59.7
	2%	DB	61.0	63.4	71.4	77.5	83.5	87.9	89.6	90.3	88.0	80.1	69.0	61.7
		MCWB	54.9	56.4	59.5	63.7	70.1	73.4	75.7	74.1	71.0	67.0	58.0	56.6
	5%	DB	57.5	59.2	66.9	74.3	81.3	85.8	87.2	87.3	84.8	76.2	64.7	57.7
		MCWB	53.6	53.6	57.0	61.6	68.8	72.4	75.0	73.6	70.4	64.1	56.8	54.2
	10%	DB	52.8	54.7	62.9	70.9	78.7	83.4	85.0	84.7	81.3	71.6	60.9	54.0
		MCWB	48.0	49.4	54.3	59.9	68.2	71.7	74.1	72.7	69.3	63.0	53.5	50.2
	0.4%	WB	59.5	62.0	64.8	68.2	74.1	77.4	80.0	78.4	75.7	73.3	63.7	61.8
		MCDB	64.1	68.7	74.8	77.4	83.0	86.7	87.8	88.1	84.3	82.0	69.7	65.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	55.8	59.4	61.7	65.9	72.2	75.9	78.0	76.4	74.1	70.2	60.7	57.9
		MCDB	59.0	63.6	69.3	74.6	80.8	84.6	86.5	84.5	83.7	76.2	65.7	61.0
	5%	WB	51.3	56.0	58.8	63.7	70.9	74.5	76.6	75.1	72.5	67.3	57.2	55.1
		MCDB	54.6	59.5	64.7	70.9	79.4	83.0	84.9	82.8	81.1	72.8	62.0	58.2
	10%	WB	46.6	51.2	55.8	61.4	69.4	73.0	75.3	73.8	71.0	63.5	53.6	51.4
		MCDB	50.3	55.5	61.1	68.8	77.3	81.2	82.8	80.9	78.4	70.8	59.1	54.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	16.4	18.0	19.3	17.8	18.2	17.2	18.2	19.4	19.0	17.6	14.5
		MCDBR	18.4	21.8	21.4	22.4	20.0	20.7	20.2	21.3	23.4	22.8	20.6	18.5
		MCWBR	16.7	17.0	14.0	13.9	10.0	9.4	8.3	8.5	9.2	10.8	13.7	15.0
	5% WB	MCDBR	18.4	20.8	18.5	19.6	17.7	18.0	16.9	16.6	18.4	18.0	17.3	16.2
		MCWBR	16.4	17.9	13.0	13.5	9.4	9.6	8.4	7.9	8.2	10.1	13.8	14.6
(40) Clear-Sky Solar Irradiance	taub		0.309	0.322	0.352	0.392	0.437	0.449	0.473	0.467	0.412	0.355	0.322	0.300
	taud		2.464	2.406	2.386	2.271	2.190	2.194	2.136	2.145	2.271	2.429	2.476	2.491
	Ebn at Noon		277	287	288	281	269	265	257	256	264	272	270	272
	Edh at Noon		26	31	35	42	46	46	49	47	39	30	25	23
	RadAvg		609	827	1173	1564	1751	1961	1853	1741	1437	1071	740	526
(45) All-Sky Solar Radiation	RadStd		66	118	107	115	162	103	104	98	149	121	82	53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+137

Nomenclature: See separate page